Evaluating Machine Learning Models for Prehospital Stroke Triage in Emergency Medical Services

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Introduction

Stroke affects over 795,000 individuals each year in the U.S., with extended time to treatment contributing to increased mortality and long-term disability[1].

1.9 Million

Neurons lost every minute during untreated stroke[1]

- Every 15 minutes = 3.6% increase in disability[1]
- Right Center = Right Treatment
- Capabilities Count
- Better Outcomes With Better Triage[2-3]

Inefficiencies in current prehospital stroke scales lead to misdiagnoses and inappropriate facility transport, resulting in costly and time-consuming interfacility transfers, which can average nearly three hours[4-8].

Objectives

- Evaluate the performance of machine learning models in detecting stroke and severe stroke using prehospital EMS data.
- Demonstrate the potential of AI to improve early triage and transport decisions by comparing model performance with standard prehospital stroke tools.

Methods

Design & Setting

Retrospective study of 4,333 patients ≥18 years old across 8,221 encounters transported by ambulance to LUMC ED between 2015 and 2020

Study Variables

- Demographics and vitals
- Cincinnati Prehospital Stroke Scale (CPSS)
- Dispatch Information
- Area Deprivation Index (ADI) by pickup location

Statistical Analysis

- Two binary classification tasks: Stroke vs non-stroke and Severe stroke vs others
- Random Forest, XGBoost, Sequential Neural Networks tuned with Optuna
- Model calibration (polynomial and temperature scaling) and feature importance (SHAP)
- Performance evaluated by ROC-AUC and PR-AUC with bootstrap CIs

Results

Figure 1. Distribution and overlap of patients based on three stroke-related criteria. Intersections involving a documented CPSS are highlighted as Subcohort 1, while the dashed outline for Subcohort 2 identifies all patients belonging to at least one of the three criteria.

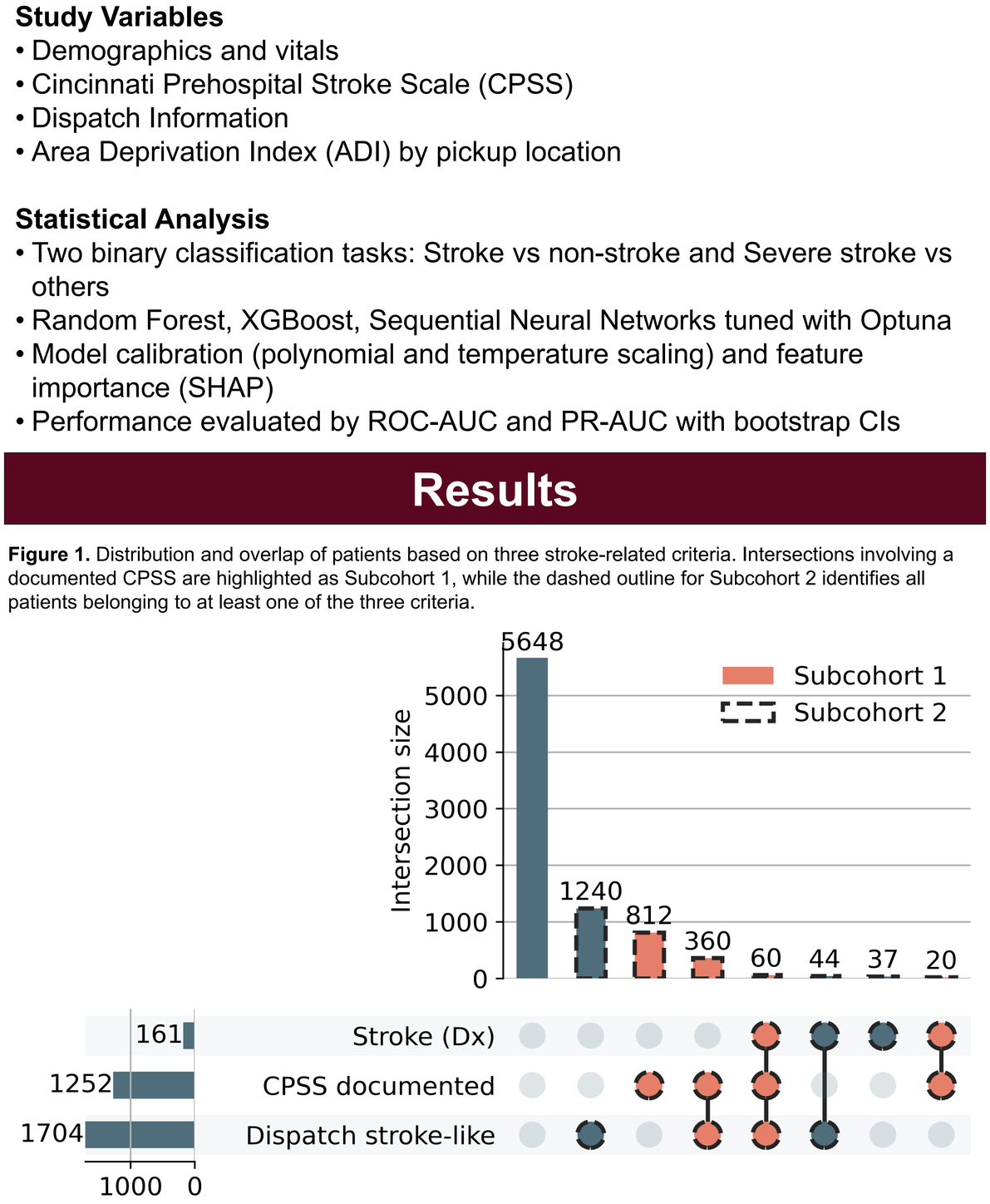
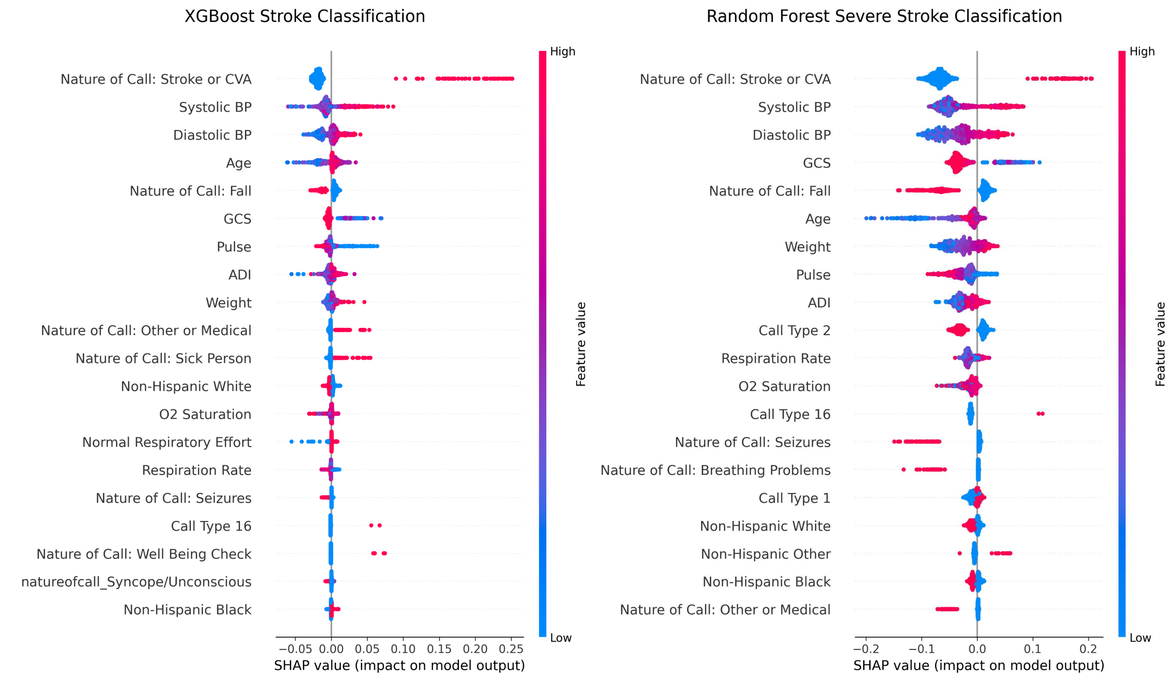


Table 1. Performance of the best models for stroke (XGBoost) and severe stroke (Random Forest) detection across three cohorts. Clinical benchmarks are included for comparison where applicable. * Indicates literature-reported performance.

	Stroke (XGBoost)					Severe Stroke (Random Forest)				
Full Cohort (n=8,221)										
Source	**Sens**	**Spec**	**PPV**	**ROC-AUC**	**PR-AUC**	**Sens**	**Spec**	**PPV**	**ROC-AUC**	**PR-AUC**
ML Model	**0.667**	**0.880**	**0.099**	**0.843**	**0.293**	**0.645**	**0.868**	**0.059**	**0.826**	**0.186**
EMT	0.323	0.982	0.263	—	—	—	—	—	—	—
Dispatch	0.453	0.984	0.360	—	—	—	—	—	—	—
EMT-Suspected Stroke Cohort (n=1,252)										
ML Model	**0.750**	**0.835**	**0.237**	**0.814**	**0.362**	**0.923**	**0.707**	**0.102**	**0.858**	**0.322**
CPSS	0.650	0.875	0.263	—	—	—	—	—	—	—
CPSS*	0.811	0.517	—	—	—	—	—	—	—	—
VAN*	—	—	—	—	—	0.810	0.380	0.290	—	—
Stroke and Mimic Cohort (n=2,573)										
ML Model	**0.688**	**0.801**	**0.186**	**0.800**	**0.330**	**0.742**	**0.760**	**0.114**	**0.811**	**0.198**

Figure 2. SHAP summary plots for the Stroke & Mimic cohort showing the top predictors for the stroke (XGBoost) and severe-stroke (Random Forest) models. Dots show feature value (color) and impact on the model output (x-axis).



Conclusion

- ML models using routine, structured EMS data detected stroke and severe stroke cases missed by EMS with more balanced sensitivity and specificity
- The Stroke & Mimic cohort best reflected real-world triage difficulty, showing that ML can aid in distinguishing strokes from plausible mimics.
- Results support AI-driven decision support as a supplement to EMS judgement, not a replacement.
- Future work: multi-site validation and richer multimodal inputs (e.g. ECG, narratives) to enhance fairness and generalizability.

References

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